ANDREW MERRITT

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Computer Science student (3.62 GPA, graduating December 2027) who builds quantitative infrastructure and autonomous AI agents in Python, Java, and C++. Ships independent projects end to end — an event-driven backtesting framework with a 126× vectorization speedup, a self-directed ML research agent, a Spring Boot/PostgreSQL trade journal — and contributes CI supply-chain hardening to SigmaHQ, the open standard for detection engineering.

EDUCATION

University of Louisville — J.B. Speed School of Engineering *Expected December 2027*

B.A. in Computer Science • GPA 3.62 / 4.0 • 84 credit hours completed • Louisville, KY

- Fall 2026 coursework: Design of Operating Systems, Introduction to Algorithms, Taxation for Business Entities. Prior: Technical Certificate, Indiana College Core — Ivy Tech Community College (GPA 3.6, 2020–2024).

TECHNICAL SKILLS

Languages Python, C++, C, Java, C#, TypeScript, JavaScript, SQL (expanding: Rust)

AI / ML PyTorch, NumPy, Pandas, Anthropic API, LLM agent orchestration and tool-use design

Systems Spring Boot, PostgreSQL, React, Node.js, REST API design, Docker, GitHub Actions CI/CD, Testcontainers, Linux

PROJECTS

axiom-backtest — Event-Driven Backtesting Framework *Python • github.com/a0merr/axiom-backtest*

- Built an event-driven backtester modeling frictions most backtests omit — per-share commission with order minimums, daily short-borrow accrual, margin interest — plus ruin detection and gross leverage caps.
- Implemented four slippage models ranked by realism (fixed, volatility-aware, square-root volume impact, order-book depth-aware), where execution cost at 50% of bar volume spans 5bp to 3,846bp under liquidity stress.
- Replaced per-bar pandas lookups with vectorized alignment: 864,332 rows/sec against 6,877, a 126× speedup processing a 10M-row dataset in ~8.5s at ~1.1GB.
- Case study under rolling-origin walk-forward validation (50/200 MA crossover, SPY/QQQ/EFA/TLT/GLD, 2010–2024): Sharpe 0.78, 95% CI [0.30, 1.27], OOS mean 0.81 across 10 folds — and reported that it underperforms buy-and-hold SPY at 0.84. Built to falsify strategies, not flatter them.

auto-research-slm / autolab — Autonomous ML Research Loop *Python • PyTorch • Anthropic API*

- Built an autonomous research agent that trains a small GPT, reads each run's loss and validation metrics, rewrites its own hyperparameters across a seven-dimension search space, and re-runs — 20 experiments per unattended overnight cycle.
- Designed the agent's tool interface and state handling so a malformed or out-of-range run result cannot corrupt the next training configuration — guardrails on an agent holding write access to its own experiment config.
- Streamed live loss curves, experiment history, and the agent's stated reasoning for every change to a browser dashboard, making each decision auditable. Companion build (autolab) runs end to end with no API key or GPU, so reviewers can reproduce it locally.

tradejournal — Full-Stack Trade Journal *Java • Spring Boot • PostgreSQL • React*

- Ingests trading-bot fills into PostgreSQL through a Spring Boot REST API, with JWT authentication and Testcontainers-backed integration tests running against a real database rather than mocks.

Terminal Arcade *Python • Standard Library Only*

- Rebuilt Snake and Tetris on a custom ANSI rendering engine with threaded input; Tetris ships a 7-bag randomizer, ghost piece, hard drop, and next-piece preview. Cross-platform, zero external dependencies.

Additional python-baseline — production-ready template (tests, CI, Docker, pre-commit) • ASList<T> generic sorted container (C++ templates) • IDD Programming Challenge 2026 (TypeScript, CI)

OPEN SOURCE

SigmaHQ / sigma — Pull Request #6080 *Under Review*

- Hardened the CI supply chain of the open detection-rule standard (4,000+ rules used across SIEM platforms) by pinning test and validation dependencies into dedicated requirements files, so an upstream release can no longer break contributor pull requests.
- Added pip caching across the GitHub Actions workflows, eliminating redundant package downloads per run. Validated on Python 3.14: full suite passing, sigma check clean, zero detection rules modified.

EXPERIENCE

Project Intern — Aqlan Lab Internship Program *May 2026 – Aug 2026*

University of Louisville • Ohio River Way, "Paddle the Ohio" water-trail initiative • Louisville, KY

- Ran field ground-truthing audits of Ohio River public access points across multiple segments, logging launch conditions, hazards, and amenities into a shared tracker, and shipped an interactive verified-improvements map (ohio-river-way-map) from the collected data.
- Built a stakeholder contact database and live dashboard coordinating outreach across partner organizations, municipalities, and community groups along the river corridor.
- Translated technical audit findings into a reusable outreach campaign pack and a 48×36 conference poster for non-technical stakeholders, converting community needs into actionable plans.

Crew Member — Taco Bell • Floyds Knobs, IN *Aug 2021 – Present*

- Entrusted with opening and closing responsibilities and peak-hour throughput while carrying a full Speed School engineering course load.

HONORS & CERTIFICATIONS

- First Year Leadership Program (top 10%) • Dean's List: Fall 2024, Fall 2025, Spring 2026 • Anthropic (17 certifications) and Google Cloud (3 badges), Jun 2026